

Association Rule Learning

DSC 340 Applied Machine Learning

EMC Ch. 5

April 14, 2025

The Plan

- Presently studying unsupervised learning (no labels!)
 - Clustering (k-Means)
 - Dimensionality Reduction (PCA)
 - Time Series (ARIMA)
- Today's topic: Association Rules (Apriori)
 - What is it? [EMC 5.1]
 - Applications [5.4]
 - Apriori algorithm [5.2]
 - Evaluating results [5.3, 5.6]
 - Helpful package [ap~~y~~ori]

Overview

EMC 5.1

Association Rule Learning

- **Unsupervised**
- Data mining problem in business (dates to 1960s)
- Input: **Transactions** in a database
 - Example: shopping cart ("**market basket analysis**")
- Output: **Rules**
 - Apocryphal example: {diapers} → {beer}
- Evaluation: Should be "**interesting**" and "**useful**"
 - Business context: should lead to insights that can be operationalized
 - For example, provide new insights into buyer behavior

Rules

- Form: $A \rightarrow B$
 - When A is observed, B is also observed (i.e., in the cart)
 - "Correlation does not imply causation" ... but in business we may not care!
- Discovery of interesting patterns
 - Note: could be used in data exploration phase of any analytics project
- Suppose we sell N items in our grocery
 - How many possible subsets are there?
- Problem: Number of rules is exponential in the number of products!
- Strategy: Prune rules that lack "support"

Support

- Threshold for inclusion in set of rules
 - AR: support for **itemset** not **class** (in classification)
- Percentage of transactions that contain an itemset L
 - Example: if 60% of all transactions contain {bread, milk} then **support** of {bread, milk} is 0.6.
- Use as criterion: minimum threshold
- Question: What are the **parameters**?
 - The rules!
- What are the **hyperparameters**?
 - Minimum threshold! Also max length of itemset. Number of iterations ...

Applications

EMC 5.4

Applications of ARs

- Better merchandizing
 - What products should be included in my inventory each month?
- Cross-merchandizing
- Physical or logical placement of product
 - Grocer could include display of diapers at the end of the beer aisle
 - Or could include a coupon for beer in the diaper aisle
 - Online shopping: "Often bought together" (Amazon)
- Promotional programs
 - loyalty card incentives, coupons mailed to customers, etc.
- Many others (recommender systems, even database sequencing!)

Apriori Algorithm

EMC 5.2

Apriori algorithm

- Agrawal and Srikant (1994)
- Bottom-up: Starts with 1-itemsets, then 2-itemsets, etc.
- Extended one at a time
- Candidates discovered each iteration
- Candidate subset added if it meets threshold
- Downward closure property
- Uses breadth-first search and Hash tree for efficient operation

Apriori(T, ϵ)

$L_1 \leftarrow \{\text{large 1-itemsets}\}$

$k \leftarrow 2$

while $L_{k-1} \neq \emptyset$

$C_k \leftarrow \{c = a \cup \{b\} \mid a \in L_{k-1} \wedge b \notin a, \{s \subseteq c \mid |s| = k-1\} \subseteq L_{k-1}\}$

for transactions $t \in T$

$D_t \leftarrow \{c \in C_k \mid c \subseteq t\}$

for candidates $c \in D_t$

$count[c] \leftarrow count[c] + 1$

$L_k \leftarrow \{c \in C_k \mid count[c] \geq \epsilon\}$

$k \leftarrow k + 1$

return $\bigcup_k L_k$

Evaluation

EMC 5.3, 5.6

Evaluation

- **Confidence:** measure of certainty or trustworthiness of discovery

$$\textit{Confidence}(X \rightarrow Y) = \frac{\textit{Support}(X \wedge Y)}{\textit{Support}(X)}$$

- **Lift:** how much more often X and Y occur together than expected

$$\textit{Lift}(X \rightarrow Y) = \frac{\textit{Support}(X \wedge Y)}{\textit{Support}(X) * \textit{Support}(Y)}$$

- **Leverage:** same idea, but uses **difference** rather than **ratio**

$$\textit{Leverage}(X \rightarrow Y) = \textit{Support}(X \wedge Y) - \textit{Support}(X) * \textit{Support}(Y)$$

Apriori at a Glance

- Set a minimum value for **support** and **confidence**
- **Extract subsets** having **support** > minimum threshold
- **Select** all **rules** from **subsets** having **confidence** > minimum threshold
- **Order** the rules by descending order of **lift**

Package

<https://pypi.org/project/apyori/>

Questions?

What are your questions?